



Project Initiation Document

Harmonising European Elevation Data Across Borders

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Abstract

Summary

1 Introduction

High-quality elevation data is increasingly available across Europe through national and regional mapping programmes. These datasets support a wide range of applications, including hydrological modelling, flood risk assessment, infrastructure planning, and environmental management. At the same time, many of these applications extend beyond administrative boundaries. The European INSPIRE framework therefore places emphasis on the interoperability of spatial datasets and on their consistent use across national borders. Elevation is one of the spatial data themes addressed within this framework (TWG-EL, 2024).

The Rhine catchment is a clear example of an area where cross-border spatial information is important. The catchment covers approximately 200,000 km² and extends across several European countries, with around 60 million people living within the basin. The Rhine and its tributaries are important for water supply, navigation, industry, agriculture, ecosystems, and flood management. Management of the river basin is already coordinated internationally through organisations such as the International Commission for the Protection of the Rhine (ICPR, 2022). Many processes within the catchment, particularly water flow and flooding, depend directly on terrain. The quality and consistency of elevation data can therefore influence analyses performed at catchment scale (Muthusamy et al., 2021).

Although detailed elevation datasets are available for large parts of Europe, they are generally collected and maintained by individual national or regional authorities. As a result, neighbouring datasets can differ in point density, spatial resolution, coordinate and height reference systems, classification, and more. Recent work by Rijkswaterstaat, TU Delft, Het Waterschapshuis, and EuroSDR documents these differences across European point cloud datasets and has identified the absence of a harmonised cross-border point cloud dataset at European scale (van der Heide et al., 2026). Such differences create practical problems when elevation data from several jurisdictions needs to be combined into a single terrain model. Simply joining the available datasets does not necessarily result in a consistent representation of the terrain.

This forms the main problem addressed by this project. The aim is to investigate how elevation data covering the Rhine catchment can be processed into a harmonised digital elevation model in an automated manner. The project considers the differences between the available source datasets and the processing steps required to bring them into a common representation. The exact geographic coverage, datasets, and harmonisation aspects considered within the project are defined further through the research questions and subquestions.

2 Literature Review

This chapter will go over the reviewed literature. It discusses both the existing approaches for harmonizing cross-border data, INSPIRE, and point cloud processing methods.

2.1 INSPIRE Specification on Elevation

In the European Union, the INSPIRE Directive (European Parliament and Council of the European Union, 2007) governs the “sharing, access, and use” of spatial data, specifically within the context of environmental policy. INSPIRE defines 34 spatial data themes; elevation data is one of those. Below is a summary of the data specification on elevation (TWG-EL, 2024), which can be considered the practical guidance on the theme. The specification contains necessary contextual information, requirements (as either general implementing rules (IR) or elevation-specific technical guidance (TG)), and recommendations.

2.1.1 INSPIRE View on Elevation

Within the scope of the specification, the elevation property — either depth or height — is understood the three-dimensional shape of the Earth’s surface (TWG-EL, 2024, p. 40–43). Generally, land-elevation and bathymetry are considered separately; for land-elevation, the elevation property is called height and its positive direction is “upwards”, and for bathymetry, it is depth with positive direction pointing downward. The scope for bathymetry is specifically the sea, inland standing water bodies, and navigable rivers. The specification also considers an integrated land-sea model, where as measured from a given datum, the height is positive and the depth is *negative*.

The specification considers two possible “shapes” of the terrain (TWG-EL, 2024, p. 43–44):

- a digital terrain model (DTM), which only represents the bare surface of the Earth;
- a digital surface model (DSM), which represents the Earth’s surface with all static natural and artificial features (e.g. this includes trees and buildings, but not cars).

“Digital elevation model” (DEM) is used as an umbrella term which encompasses both a DTM and a DSM.

2.1.2 Elevation Data Models

Within the scope of the specification, the surface is considered 2.5-dimensional — for each x, y position, only one elevation value is possible — and can be represented with gridded coverage, vector data (contour lines and spot elevations), or a triangulated irregular network (TIN). For land-elevation, **grid coverage is required**, and vector representation is recommended. For bathymetry, either grid coverage or vector representation are required.

3 Cross-border Data

4 Harmonization Pipeline

5 Conclusion

6 Recommendations

References

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